R Basic Course

Mardones, Mauricio. & Zúñiga, María José.

2025-02-11

Contents

About Course_R

Description and educational material for introductory course on R, descriptive statistics, and visualizations with applications in fisheries.

This course has these sections;

- 1. **useR**. Intro to R and basic features.
- 2. **Exploratory Data Analysis**. Basic concepts for data exploration.
- 3. **Descriptive Statistics**. Essential statistical calculations.
- 4. **Visualization**. Graphics and Maps using modern packages
- 5. **Efficient code**. Organization strategies.
- 6. **Automatization and sharing** Report generation and result sharing.
- 7. **Management of Bibliographic** sources and additional materials.

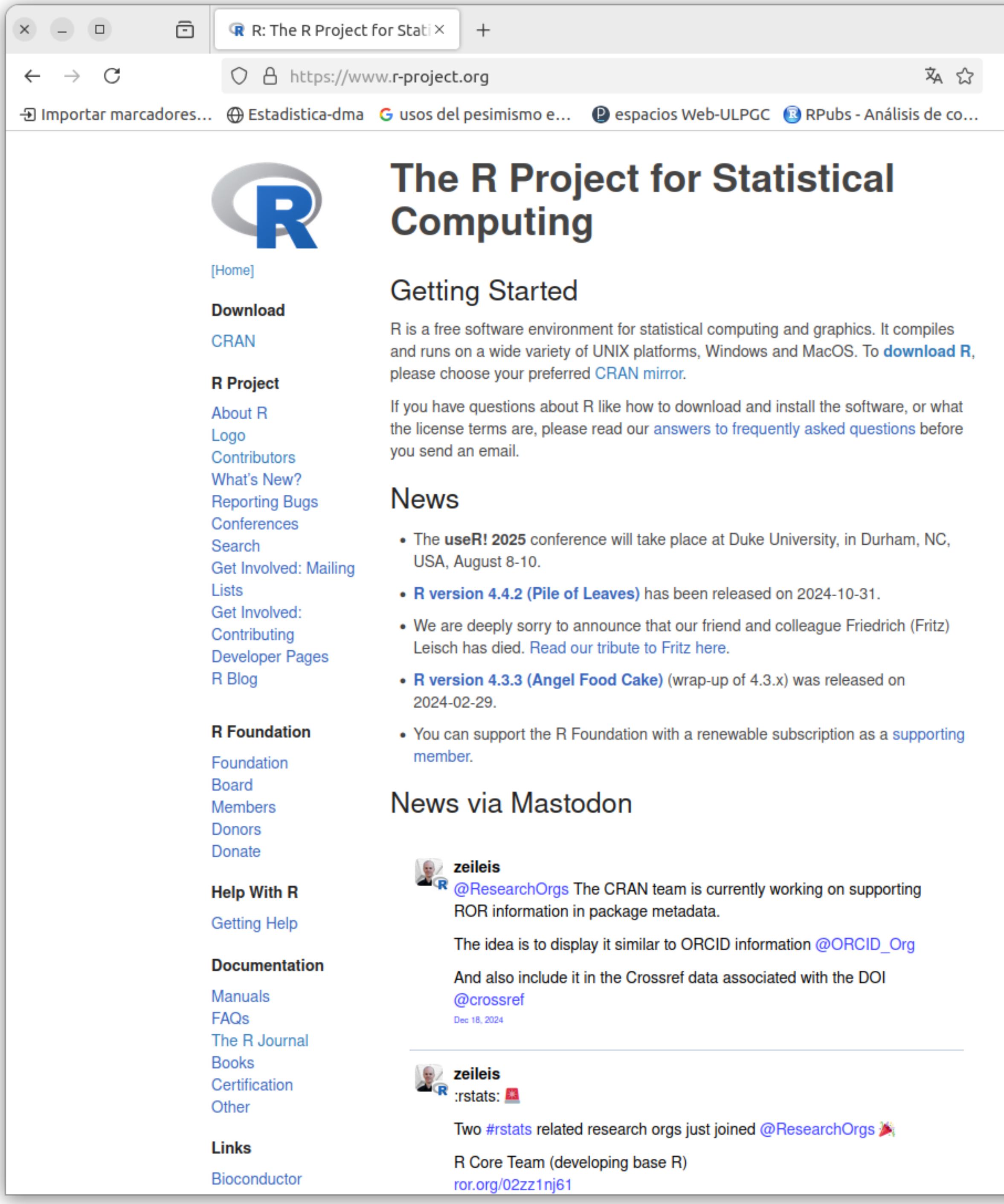
All codes and used in this course can be founded in this repository with the structure to do this Ebook Repo R Course

Chapter 1

useR. Intro to R and basic features

1.1 Welcome R user

- **R** is a programming language designed for:
  - Statistical computing
  - Data analysis
  - Visualization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If you have questions about R like how to download and install the software, or what the license terms are, please read our answers to frequently asked questions before you send an email.
News
The useR! 2025 conference will take place at Duke University, in Durham, NC, USA, August 8-10.
R version 4.4.2 (Pile of Leaves) has been released on 2024-10-31.
We are deeply sorry to announce that our friend and colleague Friedrich (Fritz) Leisch has died. Read our tribute to Fritz here.
R version 4.3.3 (Angel Food Cake) (wrap-up of 4.3.x) was released on 2024-02-29.
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@ResearchOrgs The CRAN team is currently working on supporting ROR information in package metadata.
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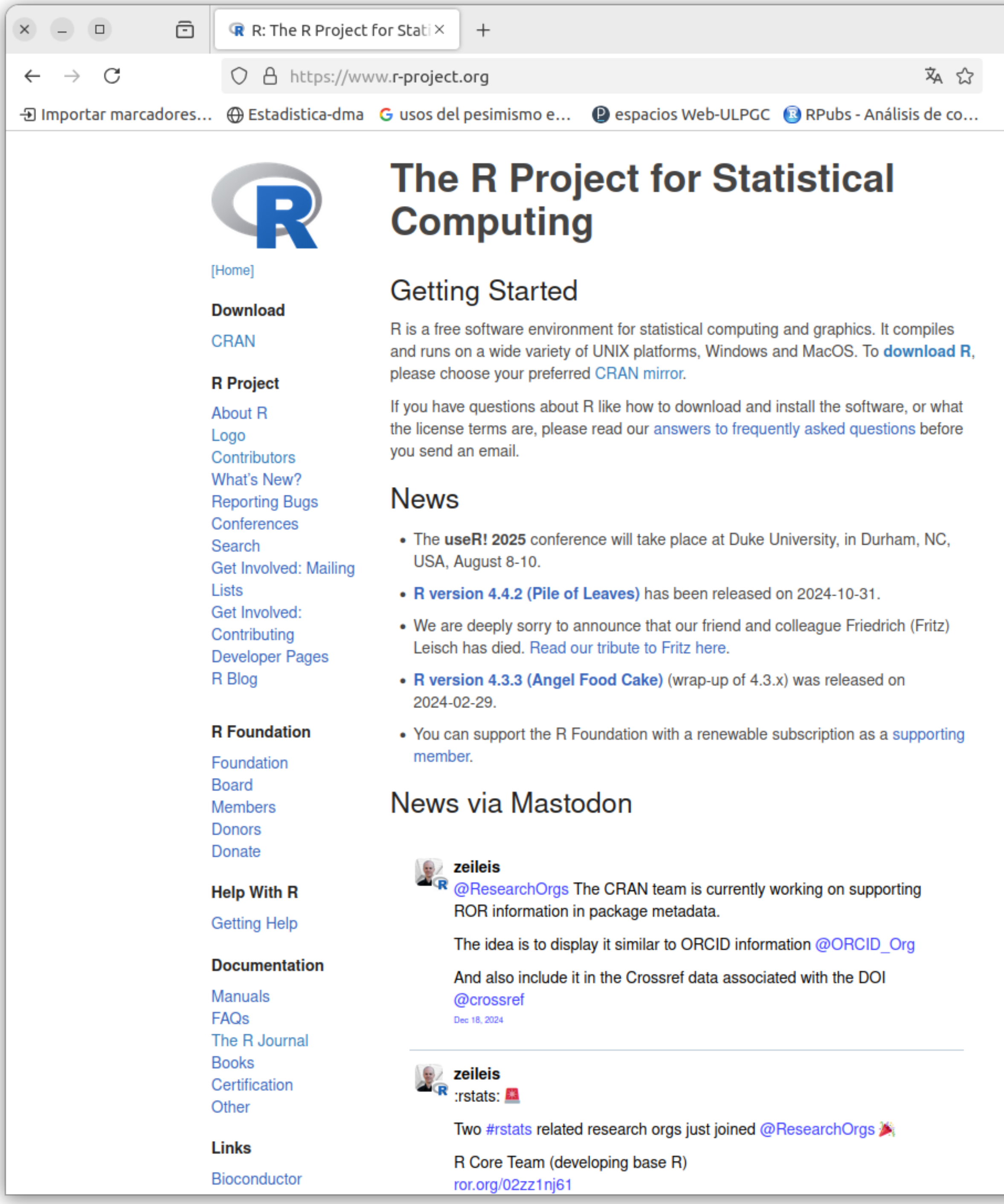
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If you have questions about R like how to download and install the software, or what the license terms are, please read our answers to frequently asked questions before you send an email.
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ror.org/02zz1nj61

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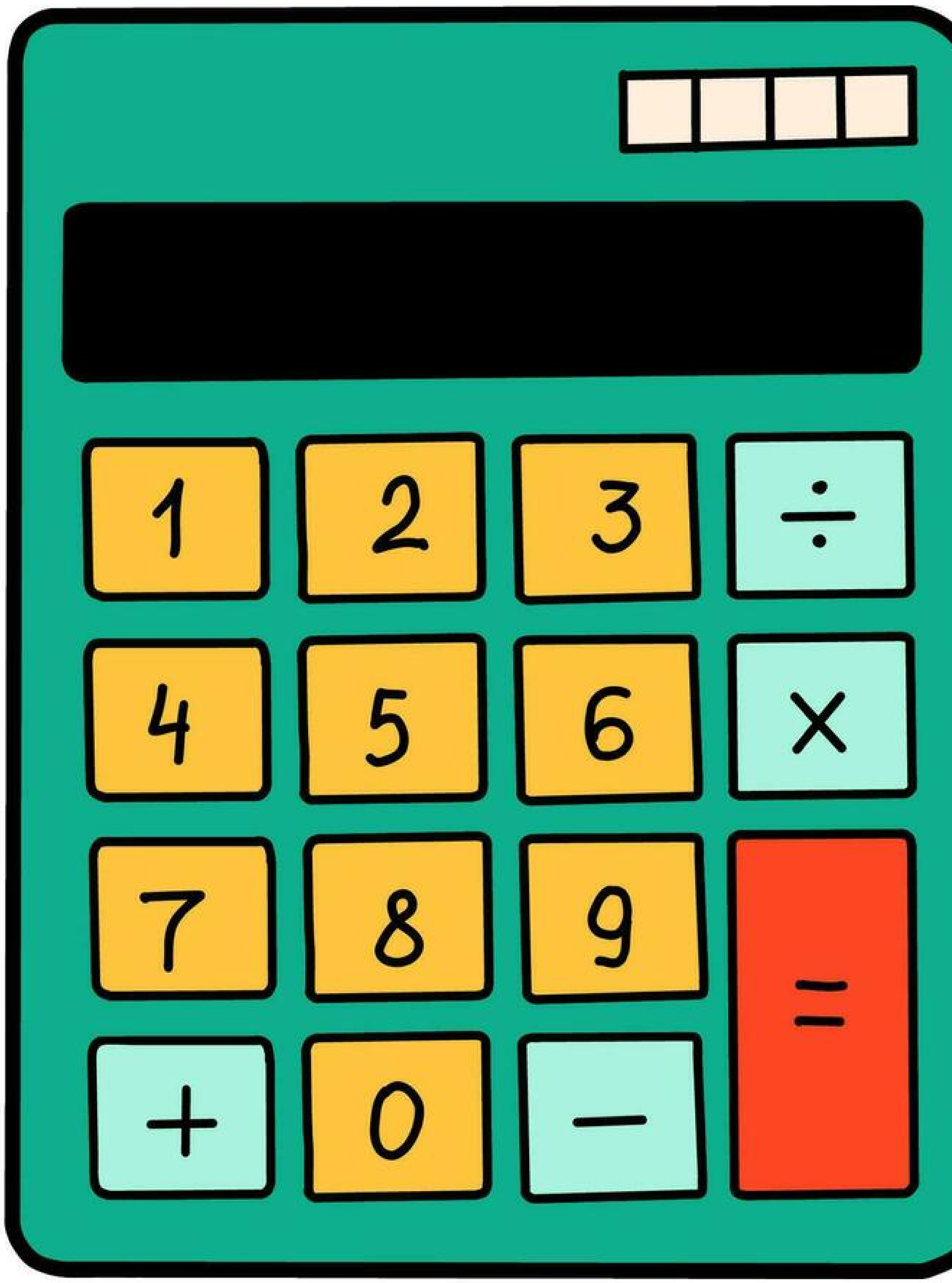
1.2 ¿Why learn R?

- Open source and free
- big community
- ~ 13.000 packages
- Reproducible science

is just this



but…

1.3 Origin of R

Seminal paper or R! here (Fig. 1.1).

1.4 Getting Started

1. Install R from CRAN
2. Install RStudio (optional, but highly recommended)
3. Write your first R command:

```
print("Hello, world!")
```

```
## [1] "Hello, world!"
```

1.5 Basic Data Types

- **Numeric:** Numbers (e.g., `3.14`, `42`)
- **Character:** Text strings (e.g., `"fish"`)
- **Logical:** `TRUE` or `FALSE`
- **Factor:** Categorical variables

1.6 Vectors

- R's most fundamental data structure:

```
numbers <- c(1, 2, 3, 4)
letters <- c("a", "b", "c")
```

- Operations are vectorized:

```
numbers * 2
```

```
## [1] 2 4 6 8
```

1.7 Data Frames

- Table-like structures:

```
df <- data.frame(Name = c("John", "Jane"),
    Age = c(25, 30))
```

- Inspect the structure:

R: A Language for Data Analysis and Graphics

Ross IHAKA and Robert GENTLEMAN

In this article we discuss our experience designing and implementing a statistical computing language. In developing this new language, we sought to combine what we felt were useful features from two existing computer languages. We feel that the new language provides advantages in the areas of portability, computational efficiency, memory management, and scoping.

Key Words: Computer language; Statistical computing.

1. INTRODUCTION

This article discusses some issues involved in the design and implementation of a computer language for statistical data analysis. Our experience with these issues occurred while developing such a language. The work has been heavily influenced by two existing languages—Becker, Chambers, and Wilks' S (1985) and Steel and Sussman's Scheme (1975). We felt that there were strong points in each of these languages and that it would be interesting to see if the strengths could be combined. The resulting language is very similar in appearance to S, but the underlying implementation and semantics are derived from Scheme. In fact, we implemented the language by first writing an interpreter for a Scheme subset and then progressively mutating it to resemble S.

We added S-like features in several stages. First, we altered the language parser so that the syntax would resemble that of S. This created a major change in the appearance of the language, but it should be emphasized that the change was entirely superficial; the underlying semantics remained those of Scheme. Next, we modified the data types of the language by removing the single scalar data type we had put into our Scheme and replacing it with the vector-based types of S. This was a much more substantive change and required major modifications to the interpreter. The final substantive change involved adding the S notion of *lazy arguments* for functions.

At this point we had enough of a framework in place to begin building a full statistical language. This process is ongoing, but we feel that we are well on the way to building a complete and useful piece of software. The development of the key portions of language

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Journal of Computational and Graphical Statistics, Volume 5, Number 3, Pages 299–314

Figure 1.1: Seminal Paper of R

```r
head(df)
```

```
##   Name Age
## 1 John  25
## 2 Jane  30
```

1.8 Visualization

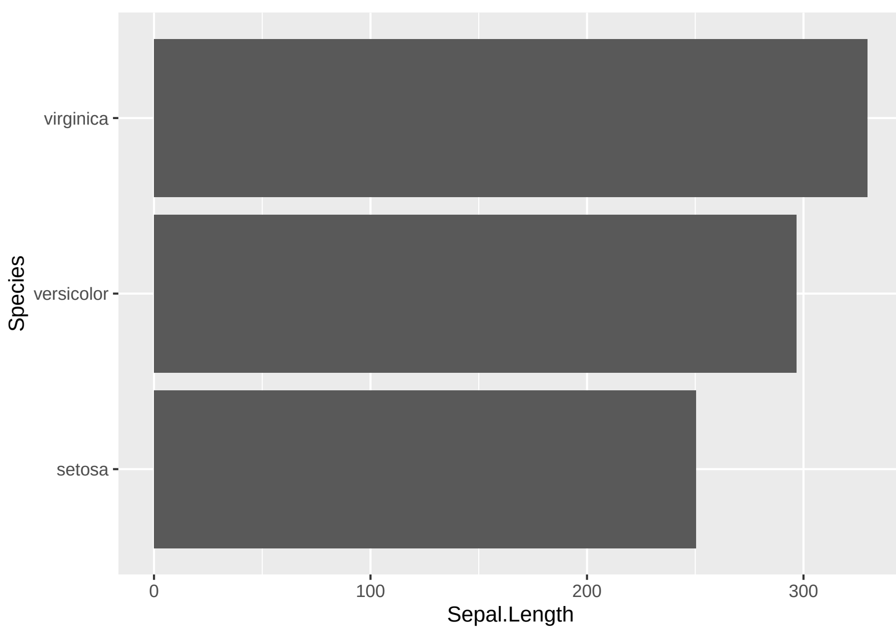
- Example base plot:

Data used

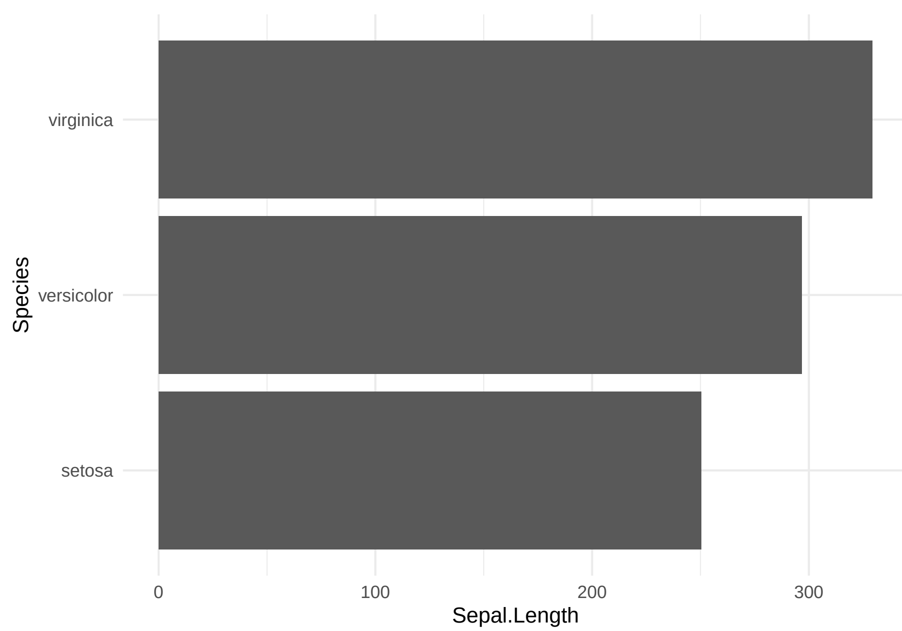
```r
kbl(head(iris))
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

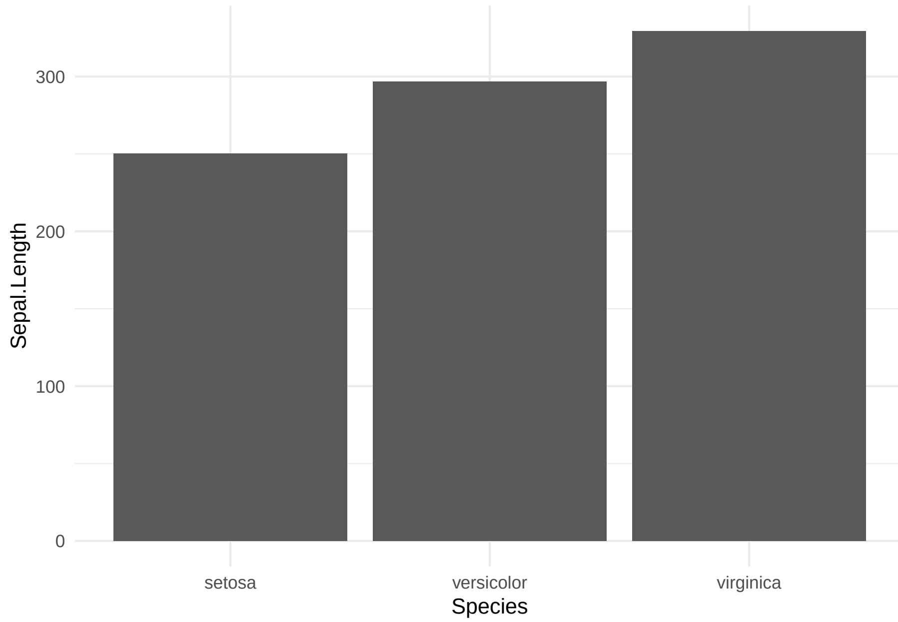
```r
plottex <- ggplot(iris, aes(x = Sepal.Length,
    y = Species)) + geom_bar(stat = "identity")
plottex
```



```
plottex <- ggplot(iris, aes(x = Sepal.Length,
    y = Species)) + geom_bar(stat = "identity") +
    theme_minimal()
plottex
```




```
plottex <- ggplot(iris, aes(x = Sepal.Length,
    y = Species)) + geom_bar(stat = "identity") +
    theme_minimal() + coord_flip()
plottex
```



1.9 Functions

- Encapsulate reusable logic:

```
square <- function(x) {
    return(x^2)
}
square(4)  # Output: 16
```

```
## [1] 16
```

1.10 Statistical Analysis

- Linear regression example:

```
fit <- lm(mpg ~ wt, data = mtcars)
```

```
summary(fit)
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
## wt           -5.3445     0.5591  -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

1.11 Conclusion

- R is powerful for:
  - Data analysis
  - Visualization
  - Statistics

1.12 Sources

Source in this link

An introduction to R

Chapter 2

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial first step in analyzing data. It provides a summary of the main characteristics of datasets and helps identify patterns, outliers, and relationships.

This section demonstrates an exploratory data analysis (EDA) of the **Palmer Penguins Dataset** using both base R and Tidyverse. It includes cleaning, descriptive statistics, and relationships between variables.

2.1 Load Necessary Packages

```
if (!require("palmerpenguins")) install.packages("palmerpenguins")
if (!require("tidyverse")) install.packages("tidyverse")

library(palmerpenguins)
library(tidyverse)
```

2.2 Load and Clean Data

```
penguins <- palmerpenguins::penguins
penguins_clean <- drop_na(penguins)
```

2.3 Basic Data Overview

```
print(head(penguins_clean))
```

```
## # A tibble: 6 x 8
```

```
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           36.7          19.3               193        3450
## 5 Adelie  Torgersen           39.3          20.6               190        3650
## 6 Adelie  Torgersen           38.9          17.8               181        3625
## # i 2 more variables: sex <fct>, year <int>
```

```r
str(penguins_clean)
```

```
## tibble [333 x 8] (S3: tbl_df/tbl/data.frame)
##  $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1
##  $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 .
##  $ bill_length_mm   : num [1:333] 39.1 39.5 40.3 36.7 39.3 38.9 39.2 41.1 38.6 34.6
##  $ bill_depth_mm    : num [1:333] 18.7 17.4 18 19.3 20.6 17.8 19.6 17.6 21.2 21.1 .
##  $ flipper_length_mm: int [1:333] 181 186 195 193 190 181 195 182 191 198 ...
##  $ body_mass_g      : int [1:333] 3750 3800 3250 3450 3650 3625 4675 3200 3800 4400
##  $ sex              : Factor w/ 2 levels "female","male": 2 1 1 1 2 1 2 1 2 2 ...
##  $ year             : int [1:333] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007
```

```r
summary(select(penguins_clean, where(is.numeric)))
```

```
##  bill_length_mm  bill_depth_mm   flipper_length_mm  body_mass_g
##  Min.   :32.10   Min.   :13.10   Min.   :172       Min.   :2700
##  1st Qu.:39.50   1st Qu.:15.60   1st Qu.:190       1st Qu.:3550
##  Median :44.50   Median :17.30   Median :197       Median :4050
##  Mean   :43.99   Mean   :17.16   Mean   :201       Mean   :4207
##  3rd Qu.:48.60   3rd Qu.:18.70   3rd Qu.:213       3rd Qu.:4775
##  Max.   :59.60   Max.   :21.50   Max.   :231       Max.   :6300
##       year
##  Min.   :2007
##  1st Qu.:2007
##  Median :2008
##  Mean   :2008
##  3rd Qu.:2009
##  Max.   :2009
```

2.4 Measures of Central Tendency

```r
mean_flipper <- mean(penguins_clean$flipper_length_mm)
median_flipper <- median(penguins_clean$flipper_length_mm)
mean_flipper
```

```
## [1] 200.967
```

```
median_flipper
```

```
## [1] 197
```

2.5 Measures of Spread

```
range_body_mass <- range(penguins_clean$body_mass_g)
var_body_mass <- var(penguins_clean$body_mass_g)
sd_body_mass <- sd(penguins_clean$body_mass_g)

cat("\nRange of body mass:", range_body_mass)
```

```
##
## Range of body mass: 2700 6300
```

```
cat("\nVariance of body mass:", var_body_mass)
```

```
##
## Variance of body mass: 648372.5
```

```
cat("\nStandard deviation of body mass:",
    sd_body_mass)
```

```
##
## Standard deviation of body mass: 805.2158
```

2.6 Frequency Tables for Categorical Data

```
cat("\nFrequency of species:\n")
```

```
##
## Frequency of species:
```

```
print(table(penguins_clean$species))
```

```
##
##    Adelie Chinstrap    Gentoo
##       146        68       119
```

2.7 Relationships Between Variables

```
cov_bill <- cov(penguins_clean$bill_length_mm,
    penguins_clean$bill_depth_mm)
cor_bill <- cor(penguins_clean$bill_length_mm,
    penguins_clean$bill_depth_mm)
```

2.8 Outlier Detection

```
q1 <- quantile(penguins_clean$body_mass_g,
    0.25)
q3 <- quantile(penguins_clean$body_mass_g,
    0.75)
iqr_value <- q3 - q1
lower_bound <- q1 - 1.5 * iqr_value
upper_bound <- q3 + 1.5 * iqr_value

outliers <- penguins_clean %>%
    filter(body_mass_g < lower_bound | body_mass_g >
        upper_bound) %>%
    select(species, body_mass_g)

cat("\nOutliers in body mass:\n")
```

```
##
## Outliers in body mass:
```

```
print(outliers)
```

```
## # A tibble: 0 x 2
## # i 2 variables: species <fct>, body_mass_g <int>
```

2.9 Comparison Between Groups

```
penguins_clean %>%
    filter(species %in% c("Adelie", "Gentoo")) %>%
    group_by(species) %>%
    summarise(mean_body_mass = mean(body_mass_g),
        sd_body_mass = sd(body_mass_g))
```

```
## # A tibble: 2 x 3
##   species mean_body_mass sd_body_mass
##   <fct>            <dbl>        <dbl>
## 1 Adelie           3706.         459.
## 2 Gentoo           5092.         501.
```

2.10 Statistical Test: t-Test

```
penguins_subset <- penguins_clean %>%
    filter(species %in% c("Adelie", "Gentoo"))
```

```
t_test_result <- t.test(body_mass_g ~ species,
    data = penguins_subset)

cat("\nT-Test Results:\n")
```

```
##
## T-Test Results:
```

```
print(t_test_result)
```

```
##
##  Welch Two Sample t-test
##
## data:  body_mass_g by species
## t = -23.254, df = 242.14, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Adelie and group Gentoo is not
## 95 percent confidence interval:
##  -1503.702 -1268.843
## sample estimates:
## mean in group Adelie mean in group Gentoo
##             3706.164             5092.437
```

The null hypothesis (`H0`) assumes that the mean body mass of *Adelie* and *Gentoo* penguins is the same. If the p-value < 0.05, we reject `H0`, indicating a significant difference in the body mass of these species.

2.11 One-Way ANOVA

```
anova_body_mass <- aov(bill_length_mm ~ species,
    data = penguins_clean)
summary(anova_body_mass)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## species       2   7015    3508   397.3 <2e-16 ***
## Residuals   330   2914       9
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2.12 Simple Linear Regression

```
model <- lm(body_mass_g ~ flipper_length_mm,
    data = penguins_clean)
summary(model)
```

```
##
```

```
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = penguins_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1057.33  -259.79   -12.24   242.97  1293.89
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -5872.09     310.29  -18.93   <2e-16 ***
## flipper_length_mm    50.15       1.54   32.56   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 393.3 on 331 degrees of freedom
## Multiple R-squared:  0.7621, Adjusted R-squared:  0.7614
## F-statistic:  1060 on 1 and 331 DF,  p-value: < 2.2e-16
```

2.13 Conclusion

This script demonstrates exploratory data analysis (EDA) using both Base R and Tidyverse, including cleaning, descriptive statistics, and relationships between variables. The statistical tests (t-test and ANOVA) provide insights into differences between groups, while regression analysis highlights potential predictive relationships.

2.14 Final Thoughts

- EDA is critical for understanding your data.
- Data handling ensures clean datasets.
- Data manipulation prepares data for in-depth analysis.
- Visualization helps reveal insights quickly.

2.15 References

Basic R

Data Explorer

EMSE 4572 CleanData

Chapter 3

Visualization.

Graphics and Maps using modern packages

3.1 Graphics

3.1.1 Base R Visualizations

Load penguins dataset

```
if (!require("palmerpenguins")) install.packages("palmerpenguins")
library(palmerpenguins)
library(data.table)
data("penguins")
```

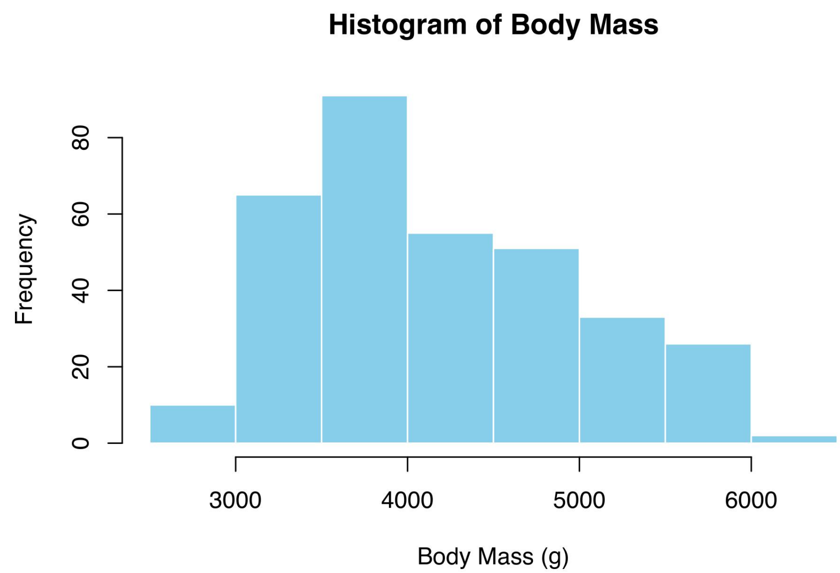
Load and Clean Data

```
penguins <- na.omit(palmerpenguins::penguins)
```

3.1.2 Visualization Techniques

Histogram of body mass

```
hist(penguins$body_mass_g, main = "Histogram of Body Mass",
    xlab = "Body Mass (g)", col = "skyblue",
    border = "white")
```

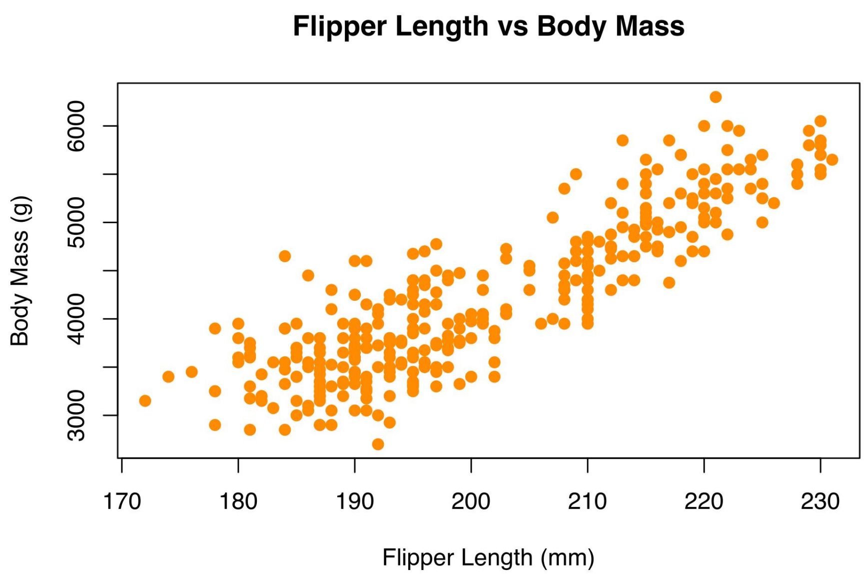


Boxplot of body mass by species

```
boxplot(body_mass_g ~ species, data = penguins,
    main = "Body Mass by Species", ylab = "Body Mass (g)",
    col = c("tomato", "lightgreen", "lightblue"))
```

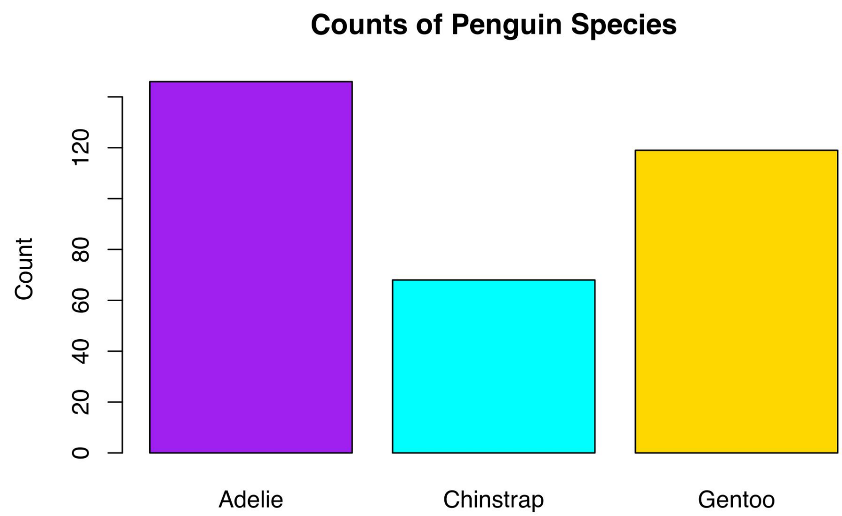
Scatter plot of flipper length vs body mass

```
plot(penguins$flipper_length_mm, penguins$body_mass_g,
    main = "Flipper Length vs Body Mass",
    xlab = "Flipper Length (mm)", ylab = "Body Mass (g)",
    pch = 19, col = "darkorange")
```



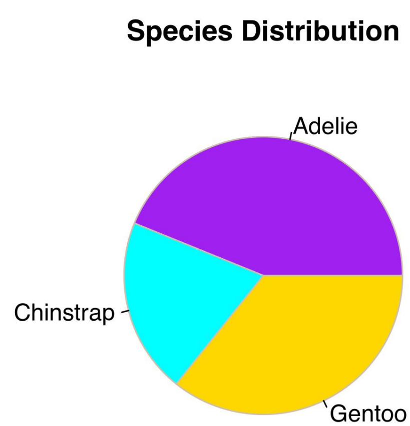
Bar plot of species counts

```
species_counts <- table(penguins$species)
barplot(species_counts, main = "Counts of Penguin Species",
    col = c("purple", "cyan", "gold"), ylab = "Count")
```



Pie chart of species distribution

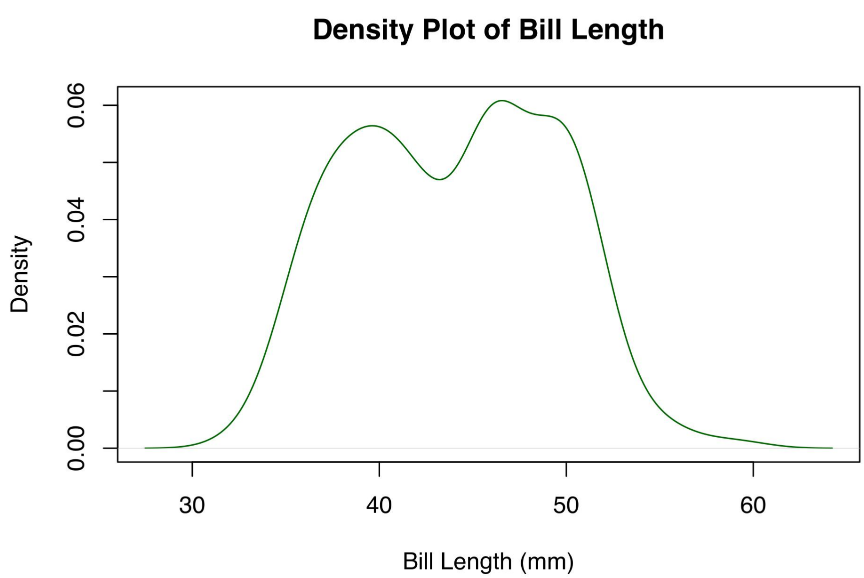
```
pie(species_counts, main = "Species Distribution",
    col = c("purple", "cyan", "gold"), border = "antiquewhite3")
```



```
# cl <- colors() Nº colores
```

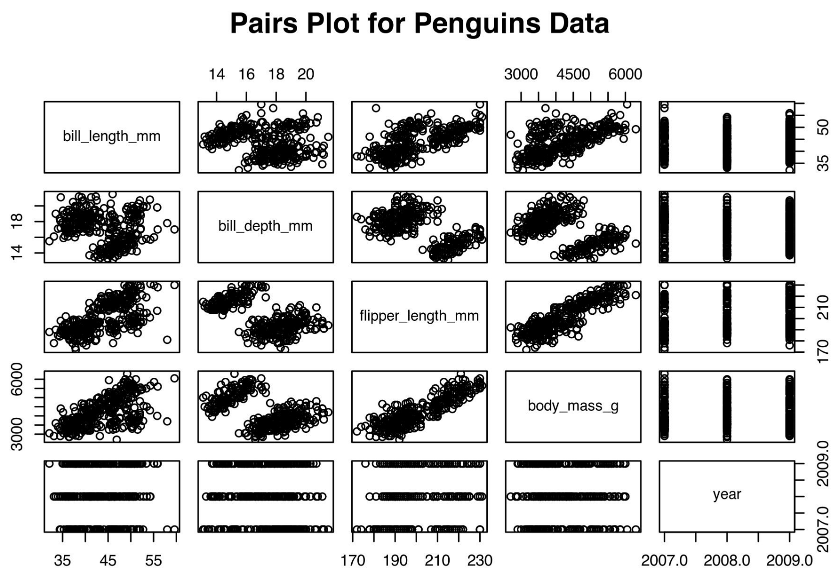
Density plot of bill length

```
plot(density(penguins$bill_length_mm), main = "Density Plot of Bill Length",
    xlab = "Bill Length (mm)", col = "darkgreen")
```



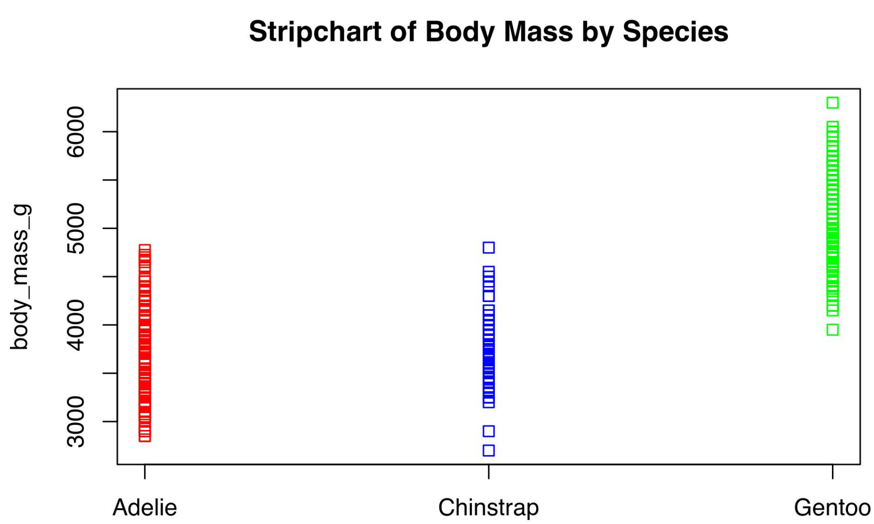
Pairs plot for numeric variables

```
pairs(penguins[, sapply(penguins, is.numeric)],
    main = "Pairs Plot for Penguins Data")
```



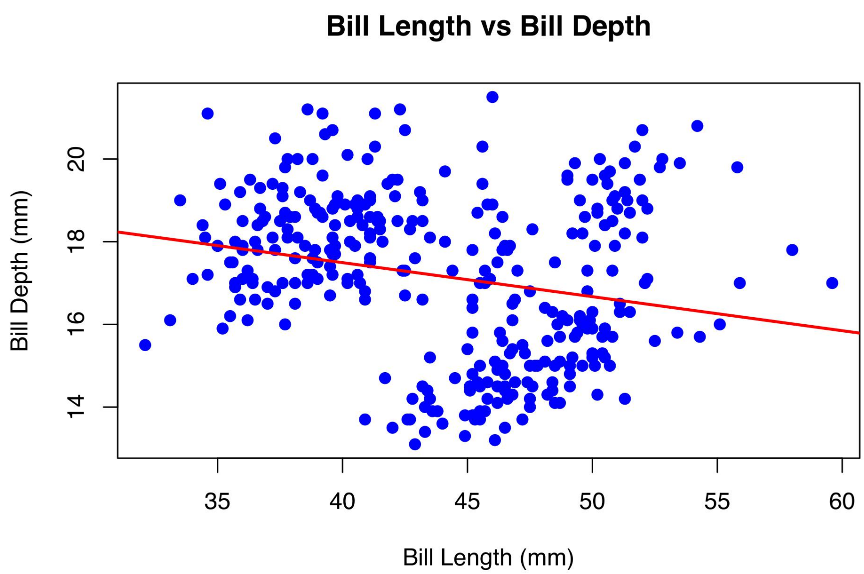
Stripchart of body mass by species

```
stripchart(body_mass_g ~ species, data = penguins,
    vertical = TRUE, main = "Stripchart of Body Mass by Species",
    col = c("red", "blue", "green"))
```



Scatter plot with regression line

```
plot(penguins$bill_length_mm, penguins$bill_depth_mm,
    main = "Bill Length vs Bill Depth", xlab = "Bill Length (mm)",
    ylab = "Bill Depth (mm)", pch = 19, col = "blue")
abline(lm(bill_depth_mm ~ bill_length_mm,
    data = penguins), col = "red", lwd = 2)
```



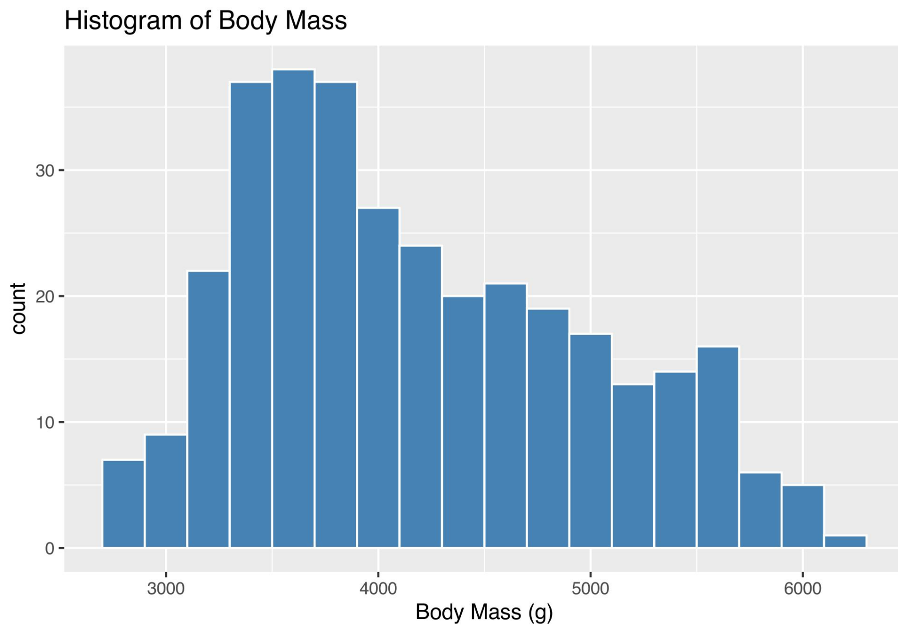
3.2 Tidyverse universe Visualizations

Load necessary package

```
if (!require("tidyverse")) install.packages("tidyverse")
library(tidyverse)
```

Histogram using ggplot2

```
ggplot(penguins, aes(x = body_mass_g)) +
    geom_histogram(binwidth = 200, fill = "steelblue",
        color = "white") + labs(title = "Histogram of Body Mass",
    x = "Body Mass (g)")
```

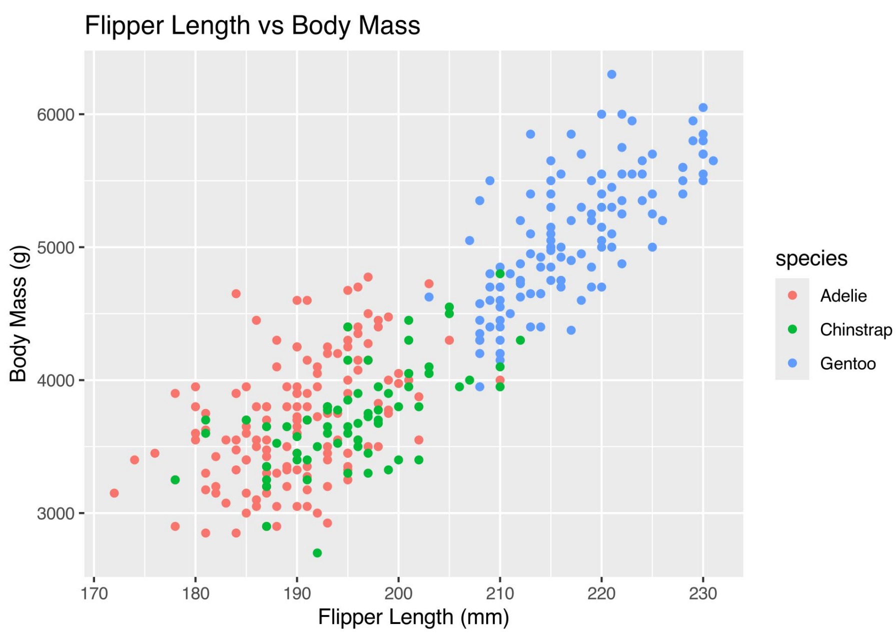


Boxplot of body mass by species

```
body <- ggplot(penguins, aes(x = species,
    y = body_mass_g, fill = species)) + geom_boxplot() +
    labs(title = "Body Mass by Species",
        y = "Body Mass (g)")
```

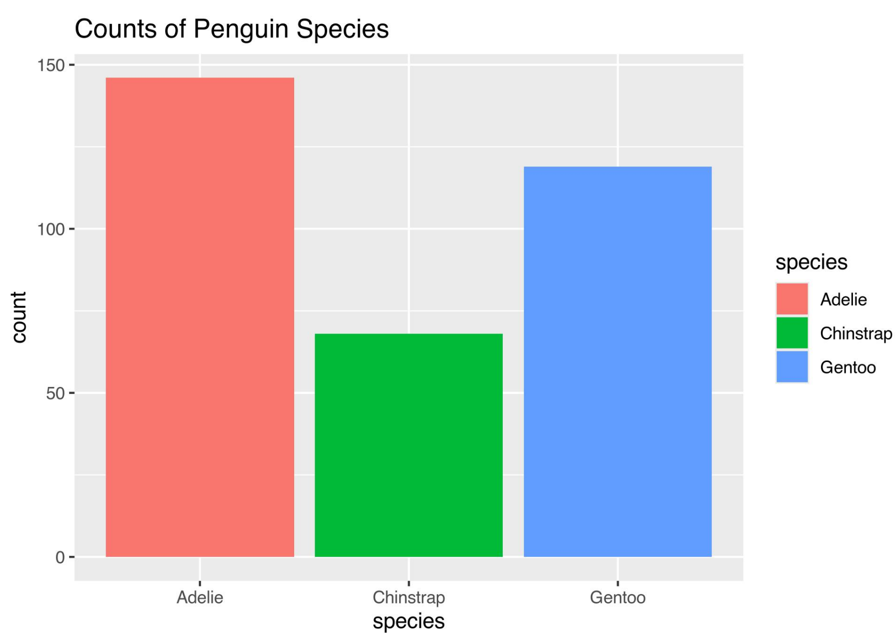
Scatter plot of flipper length vs body mass

```
ggplot(penguins, aes(x = flipper_length_mm,
    y = body_mass_g, color = species)) +
    geom_point() + labs(title = "Flipper Length vs Body Mass",
    x = "Flipper Length (mm)", y = "Body Mass (g)")
```



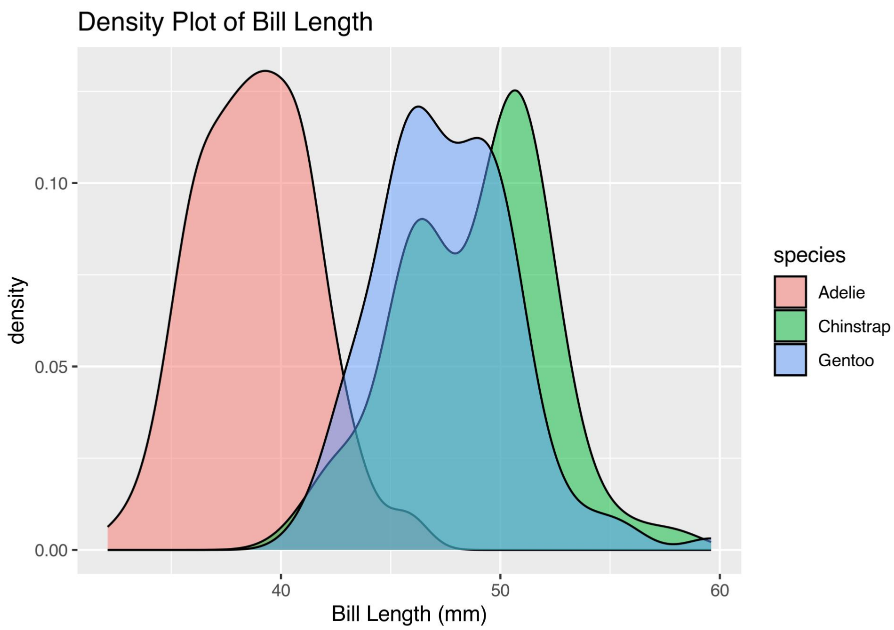
Bar plot of species counts

```
ggplot(penguins, aes(x = species, fill = species)) +
    geom_bar() + labs(title = "Counts of Penguin Species")
```



Density plot of bill length

```
ggplot(penguins, aes(x = bill_length_mm,
    fill = species)) + geom_density(alpha = 0.5) +
    labs(title = "Density Plot of Bill Length",
        x = "Bill Length (mm)")
```

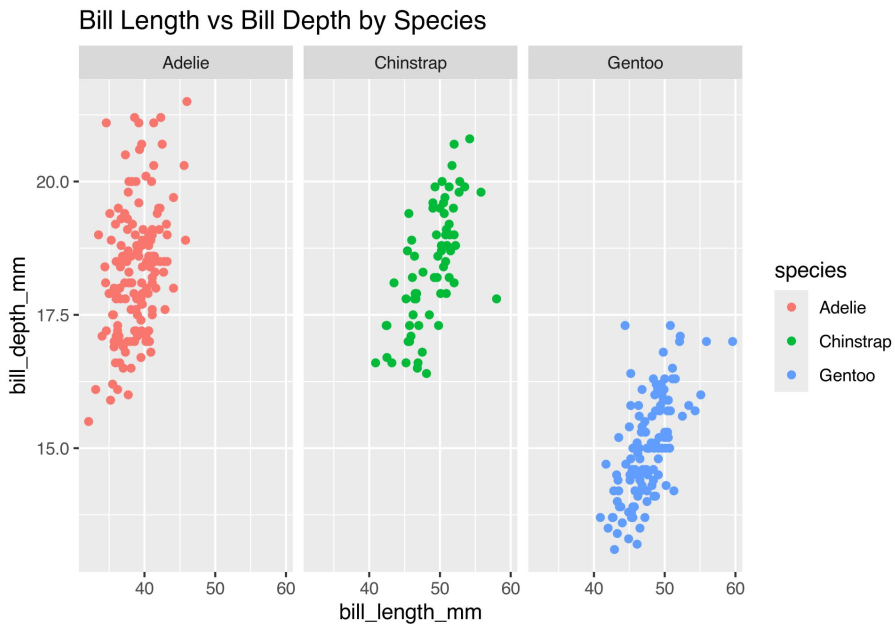


Violin plot of body mass by species

```
ggplot(penguins, aes(x = species, y = body_mass_g,
    fill = species)) + geom_violin() + labs(title = "Violin Plot of Body Mass by Speci
    y = "Body Mass (g)")
```

Faceted scatter plot

```
ggplot(penguins, aes(x = bill_length_mm,
    y = bill_depth_mm, color = species)) +
    geom_point() + facet_wrap(~species) +
    labs(title = "Bill Length vs Bill Depth by Species")
```



Correlations plot

Select numeric columns and compute correlation matrix

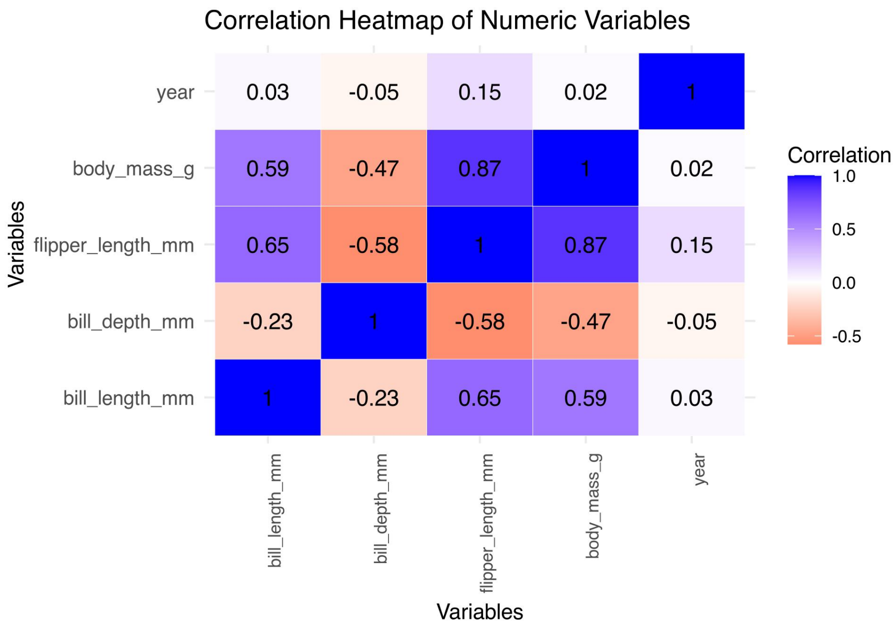
```
penguins_numeric <- penguins %>%
    select(where(is.numeric))
cor_matrix <- cor(penguins_numeric, use = "complete.obs")
```

Melt the correlation matrix for ggplot

```
cor_data <- data.table::melt(cor_matrix,
    varnames = c("Var1", "Var2"))
```

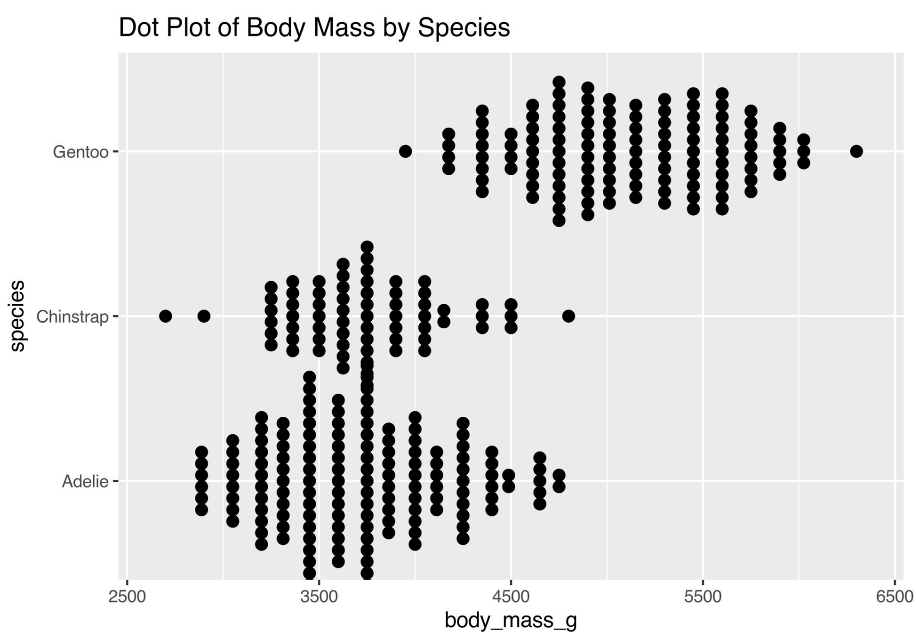
Create the heatmap plot

```
ggplot(cor_data, aes(x = Var1, y = Var2,
    fill = value)) + geom_tile(color = "white") +
    scale_fill_gradient2(low = "red", mid = "white",
        high = "blue", midpoint = 0) + geom_text(aes(label = round(value,
    2)), color = "black", size = 4) + theme_minimal() +
    theme(axis.text.x = element_text(angle = 90,
        hjust = 1), axis.text.y = element_text(size = 10)) +
    labs(title = "Correlation Heatmap of Numeric Variables",
        x = "Variables", y = "Variables",
        fill = "Correlation")
```



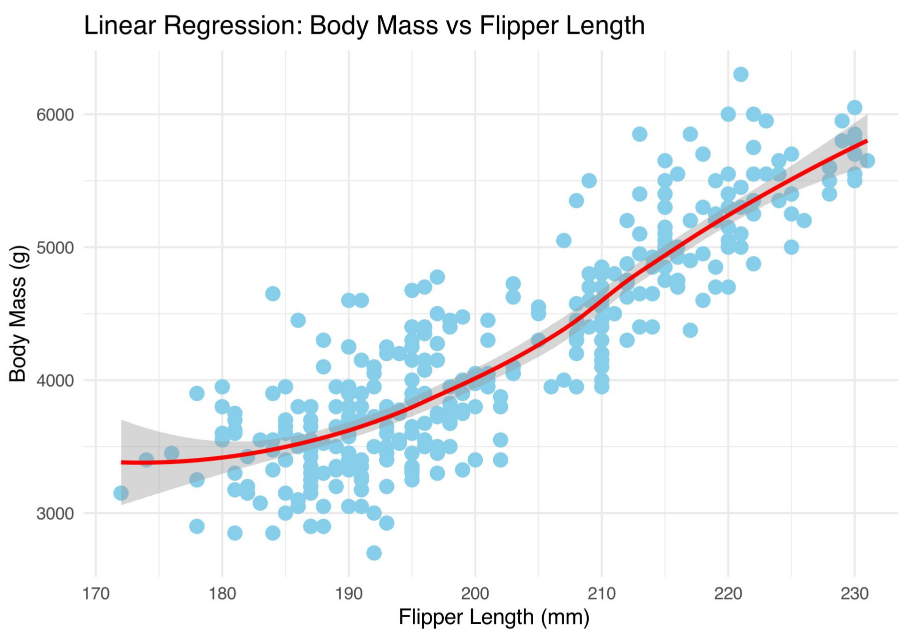
Dot plot

```r
ggplot(penguins, aes(x = body_mass_g, y = species)) +
    geom_dotplot(binaxis = "x", stackdir = "center",
        dotsize = 0.5) + labs(title = "Dot Plot of Body Mass by Species")
```



Plot regression with tidyverse

```
penguins %>%
    ggplot(aes(x = flipper_length_mm, y = body_mass_g)) +
    geom_point(color = "skyblue", size = 3) +
    geom_smooth(method = "loess", color = "red",
        se = TRUE) + labs(title = "Linear Regression: Body Mass vs Flipper Length",
    x = "Flipper Length (mm)", y = "Body Mass (g)") +
    theme_minimal()
```



Multivariate Analysis Example (PCA)

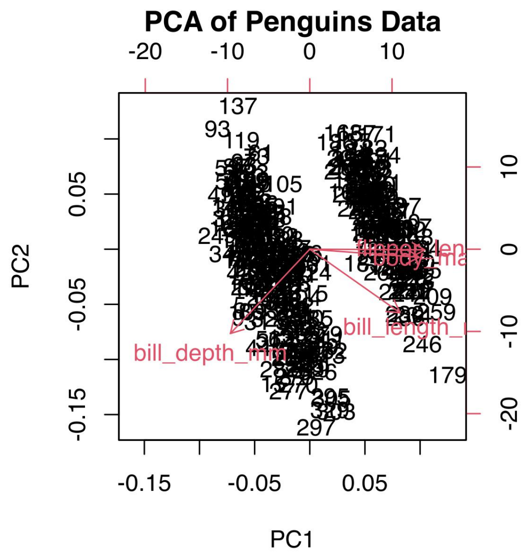
Base R PCA

```
pca_data <- penguins %>%
    select(c("bill_length_mm", "bill_depth_mm",
        "flipper_length_mm", "body_mass_g"))
pca_result <- prcomp(na.omit(pca_data), scale. = TRUE)
summary(pca_result)
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4
## Standard deviation     1.6569 0.8821 0.60716 0.32846
## Proportion of Variance 0.6863 0.1945 0.09216 0.02697
## Cumulative Proportion  0.6863 0.8809 0.97303 1.00000
```

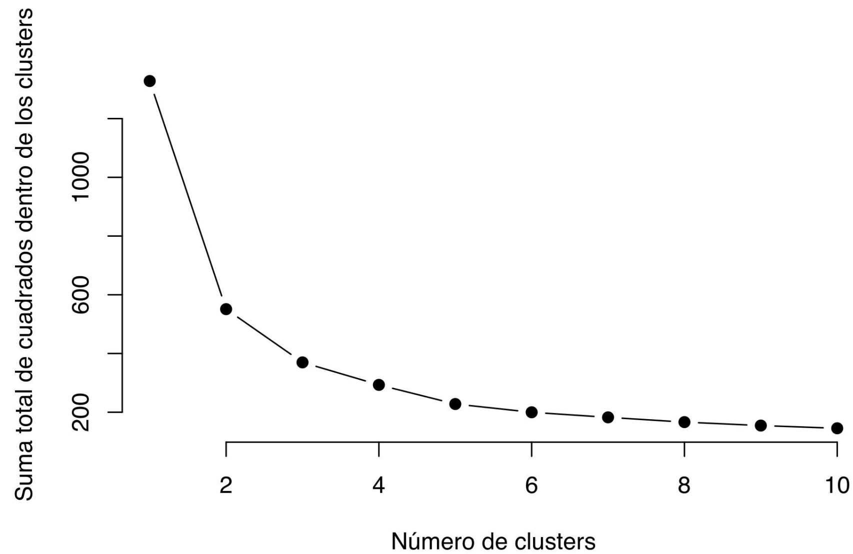
PCA Plot in Base R

```r
biplot(pca_result, main = "PCA of Penguins Data")
```

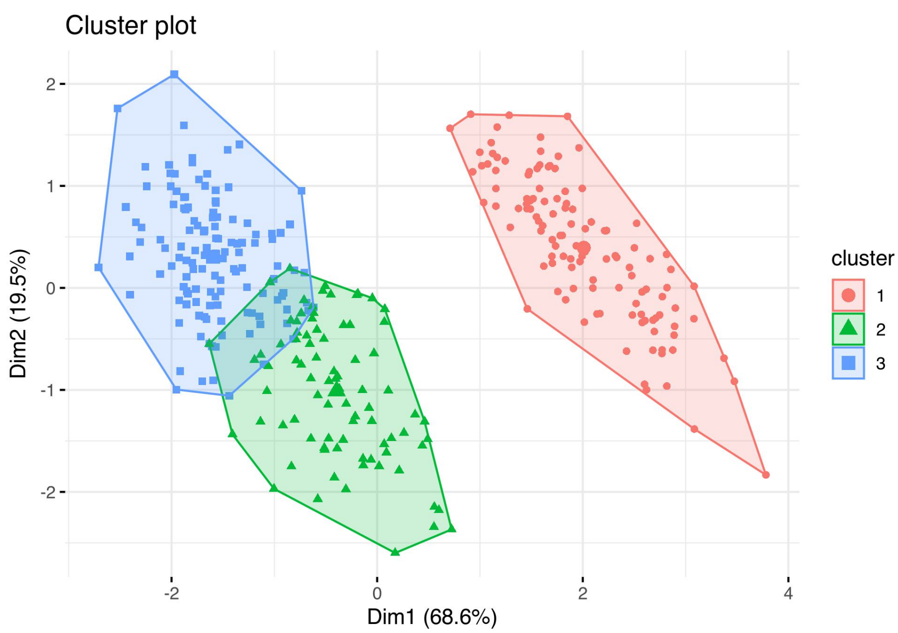


Clustering with k-means

```r
library(cluster)

# Seleccionar solo las columnas
# numéricas necesarias (removiendo NAs)
data <- na.omit(penguins[, c("bill_length_mm",
    "bill_depth_mm", "flipper_length_mm",
    "body_mass_g")])

# Escalado de los datos
scaled_data <- scale(data)

# Selección del número óptimo de
# clústeres (Método Elbow)
wss <- sapply(1:10, function(k) {
    kmeans(scaled_data, centers = k, nstart = 25)$tot.withinss
})

# Gráfico del codo
plot(1:10, wss, type = "b", pch = 19, frame = FALSE,
    xlab = "Número de clusters", ylab = "Suma total de cuadrados dentro de los clusters")
```




```r
# Aplicación del algoritmo k-means
kmeans_result <- kmeans(scaled_data, centers = 3,
    nstart = 25)
data$cluster <- as.factor(kmeans_result$cluster)

# Visualización de clústeres
library(factoextra)

# Visualización en un espacio PCA
fviz_cluster(kmeans_result, data = scaled_data,
    geom = "point", ellipse.type = "convex",
    ggtheme = theme_minimal())
```



3.3 Simple maps

Instalar paquetes necesarios si no están instalados.

```
library(sf)
library(ggplot2)
library(dplyr)
library(ggspatial)
library(rnaturalearth)
library(rnaturalearthdata)
library(terra)
library(lubridate)
```

Chile example:

```
chile <- ne_states(country = "Chile", returnclass = "sf")
```

Puerto Montt example

```
region_puerto_montt <- chile %>%
    filter(name_en == "Los Lagos")
```

random coordinates with variable

```
lat_center <- -41.469
lon_center <- -72.942
```

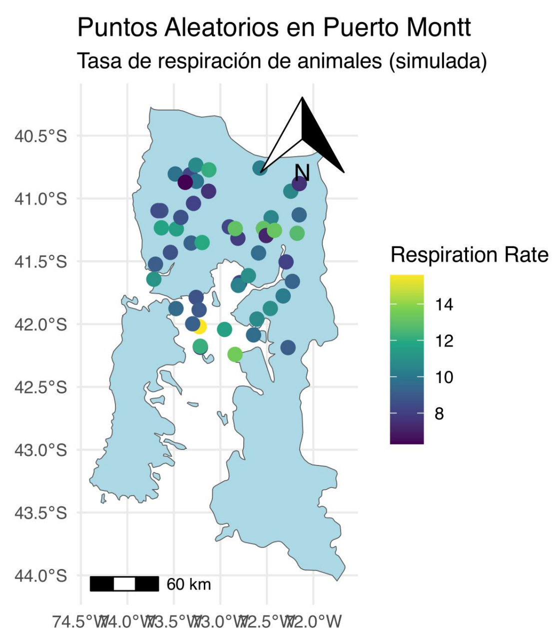
```
# set.seed(42) # Semilla para
# reproducibilidad
n_points <- 50
points_data <- data.frame(lon = runif(n_points,
    lon_center - 0.8, lon_center + 0.8),
    lat = runif(n_points, lat_center - 0.8,
        lat_center + 0.8), respiration_rate = rnorm(n_points,
        mean = 10, sd = 2))
```

Convert `sf` object

```
points_sf <- st_as_sf(points_data, coords = c("lon",
    "lat"), crs = 4326)
```

Test map with attributes

```
plot <- ggplot(data = region_puerto_montt) +
    geom_sf(fill = "lightblue") + geom_sf(data = points_sf,
    aes(color = respiration_rate), size = 3) +
    scale_color_viridis_c() + annotation_scale(location = "bl") +
    annotation_north_arrow(location = "tr",
        which_north = "true") + labs(title = "Puntos Aleatorios en Puerto Montt",
    subtitle = "Tasa de respiración de animales (simulada)",
    color = "Respiration Rate") + theme_minimal()

print(plot)
```



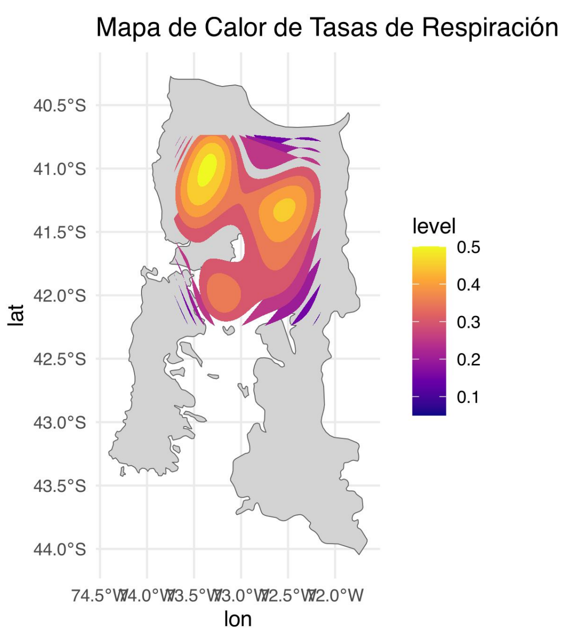
Convertir a objeto espacial sf

```
points_sf <- st_as_sf(points_data, coords = c("lon",
    "lat"), crs = 4326)
```

3.3.1 Other maps

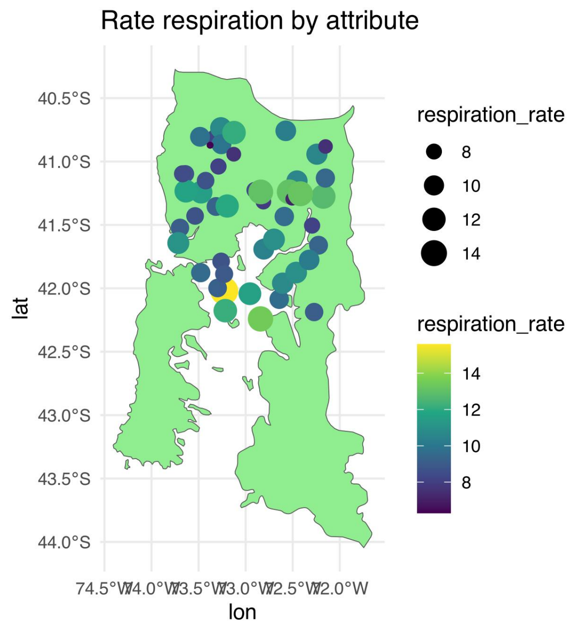
Heat map

```
ggplot(data = region_puerto_montt) + geom_sf(fill = "lightgrey") +
    stat_density_2d(data = points_data, aes(x = lon,
        y = lat, fill = ..level..), geom = "polygon") +
    scale_fill_viridis_c(option = "C") +
    labs(title = "Mapa de Calor de Tasas de Respiración") +
    theme_minimal()
```



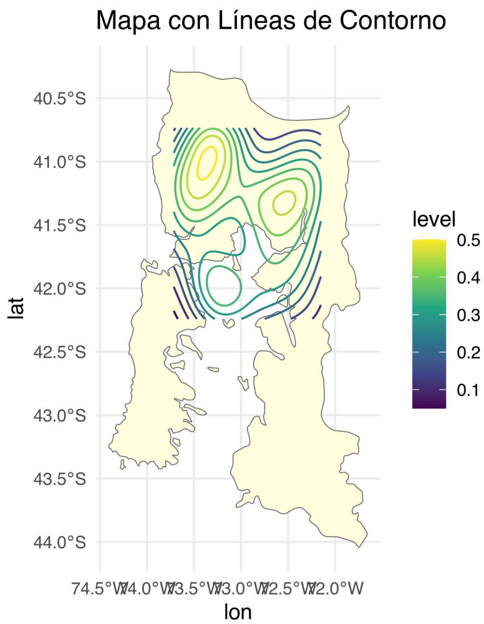
Points by attribute

```
ggplot(data = region_puerto_montt) + geom_sf(fill = "lightgreen") +
    geom_point(data = points_data, aes(x = lon,
        y = lat, size = respiration_rate,
        color = respiration_rate)) + scale_color_viridis_c() +
    labs(title = "Rate respiration by attribute") +
    theme_minimal()
```



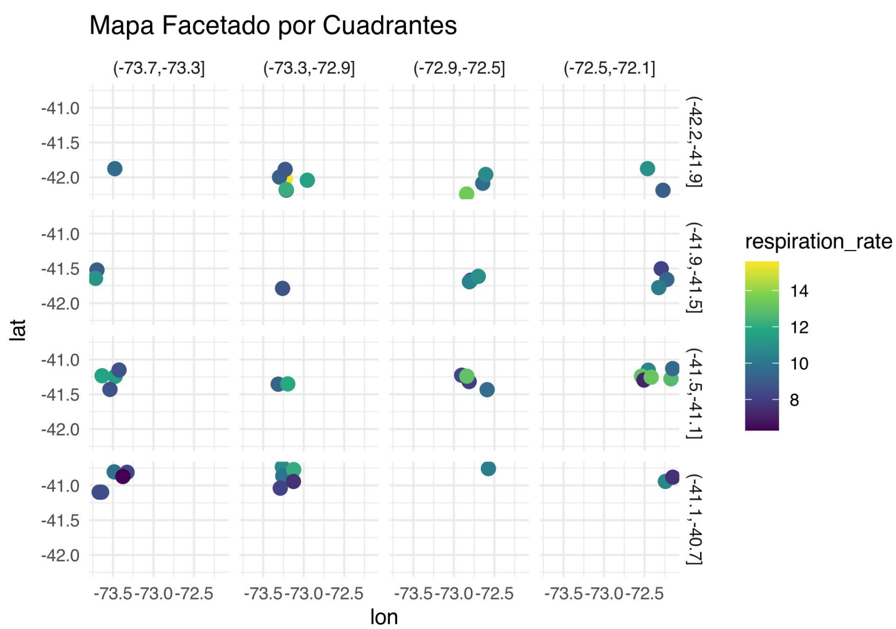
Mapa con líneas de contorno

```
ggplot(data = region_puerto_montt) + geom_sf(fill = "lightyellow") +
    stat_density_2d(data = points_data, aes(x = lon,
        y = lat, color = ..level..), geom = "density2d") +
    scale_color_viridis_c() + labs(title = "Mapa con Líneas de Contorno") +
    theme_minimal()
```



Mapa por cuadrantes

```
points_data <- points_data %>%
    mutate(lat_group = cut(lat, breaks = 4),
        lon_group = cut(lon, breaks = 4))

ggplot(points_data, aes(x = lon, y = lat,
    color = respiration_rate)) + geom_point(size = 3) +
    facet_grid(lat_group ~ lon_group) + scale_color_viridis_c() +
    labs(title = "Mapa Facetado por Cuadrantes") +
    theme_minimal()
```




```
# Cargar shapefile en R ruta_shapefile
# <- 'ruta/del/archivo.shp'
# shapefile_data <-
# st_read(ruta_shapefile)
# plot(st_geometry(shapefile_data))
```

```
# Cargar archivo Excel con datos
# georreferenciados ruta_excel <-
# 'ruta/del/archivo.xlsx' datos_excel
# <- read_excel(ruta_excel) sf_datos <-
# st_as_sf(datos_excel, coords =
# c('lon', 'lat'), crs = 4326) ggplot()
# + geom_sf(data = region_puerto_montt,
# fill = 'lightgrey') + geom_sf(data =
# sf_datos, color = 'red', size = 3) +
# theme_minimal()
```

The `sf` package (Simple Features) by Edzer Pebesma is a crucial tool for spatial data analysis in R. It provides a modern, efficient, and standardized way to handle vector spatial data using the Simple Features specification defined by the Open Geospatial Consortium (OGC). Here's why it's significant and why it has become essential compared to older, deprecated libraries:

3.3.2 Key Advantages of `sf`:

1. **Simple Features Standard:**

   - Adheres to the OGC Simple Features standard, which ensures compatibility with modern GIS systems (e.g., QGIS, PostGIS).
2. **Enhanced Performance:**
   - Operates on top of the `GEOS`, `GDAL`, and `PROJ` libraries, offering high-performance spatial operations.

   - More memory-efficient than older alternatives like `sp`.
3. **Easy Integration with the Tidyverse:**
   - `sf` objects (data frames with a geometry column) integrate seamlessly with `dplyr`, `ggplot2`, and other tidyverse packages.
4. **Better Handling of Coordinate Systems:**
   - Simplifies the manipulation and transformation of coordinate reference systems (CRS).
   - Easily handles projections via the `st_crs()` function.
5. **Deprecated Libraries:**
   - Older packages like `sp` and `rgdal` have been phased out due to their complex S4 structure and lack of maintenance.

   - `sf` uses a simpler and more flexible approach with class `sfc` (simple feature column), making operations straightforward.

3.3.3 Useful Functions

- `st_read()`: Import spatial data (shapefiles, geoJSON, etc.).
- `st_transform()`: Change CRS.
- `st_intersection()`, `st_union()`: Spatial operations.
- `ggplot2::geom_sf()`: Plot spatial data directly.

3.3.4 Additional Resources

- Official `sf` Documentation

- CRAN Page for `sf`

- Simple Features Standard by OGC

- R-spatial Blog for spatial data tutorials

Chapter 4

Efficient Code

4.1 Equations

Here is an equation.

$$f\left(k\right) = \binom{n}{k} p^k \left(1-p\right)^{n-k} \tag{4.1}$$

You may refer to using `\@ref(eq:binom)`, like see Equation (4.1).

Read more here https://bookdown.org/yihui/bookdown/markdown-extensions-by-bookdown.html.

4.2 Callout blocks

The R Markdown Cookbook provides more help on how to use custom blocks to design your own callouts: https://bookdown.org/yihui/rmarkdown-cookbook/custom-blocks.html

Chapter 5

Reporting and Sharing

5.1 Publishing

HTML books can be published online, see: https://bookdown.org/yihui/bookdown/publishing.html

5.2 404 pages

By default, users will be directed to a 404 page if they try to access a webpage that cannot be found. If you'd like to customize your 404 page instead of using the default, you may add either a `_404.Rmd` or `_404.md` file to your project root and use code and/or Markdown syntax.

5.3 Metadata for sharing

Bookdown HTML books will provide HTML metadata for social sharing on platforms like Twitter, Facebook, and LinkedIn, using information you provide in the `index.Rmd` YAML. To setup, set the `url` for your book and the path to your `cover-image` file. Your book's `title` and `description` are also used.

This `gitbook` uses the same social sharing data across all chapters in your book-all links shared will look the same.

Specify your book's source repository on GitHub using the `edit` key under the configuration options in the `_output.yml` file, which allows users to suggest an edit by linking to a chapter's source file.

Read more about the features of this output format here:

https://pkgs.rstudio.com/bookdown/reference/gitbook.html

Or use:

```
`?`(bookdown::gitbook)
```